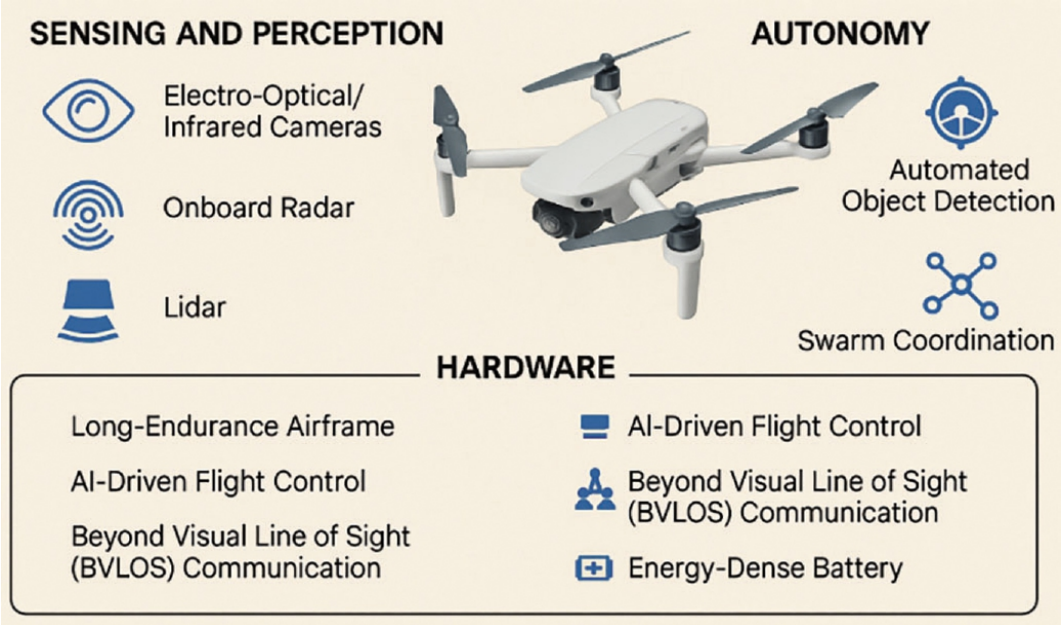


Figure 5.1 Fourth-generation drones.

5.4 Swarm Technology and Collaborative Drones

5.4.1 Introduction to swarm intelligence and group coordination

Swarm intelligence refers to the collective behavior of multiple autonomous drones that interact locally with one another and with their environment to achieve global objectives efficiently. Each drone in the swarm is equipped with sensors, communication modules, and onboard processors that allow it to perceive surroundings, share situational data, and adjust behavior in real time. Unlike traditional UAV systems dependent on centralized control, swarms use distributed decision-making algorithms such as boids algorithms, particle swarm optimization (PSO), and behavior-based coordination to self-organize and accomplish complex missions cooperatively. The core principles of swarm coordination include:

- *Decentralization*: Each unit operates independently, contributing to collective decisions without needing a central command.